

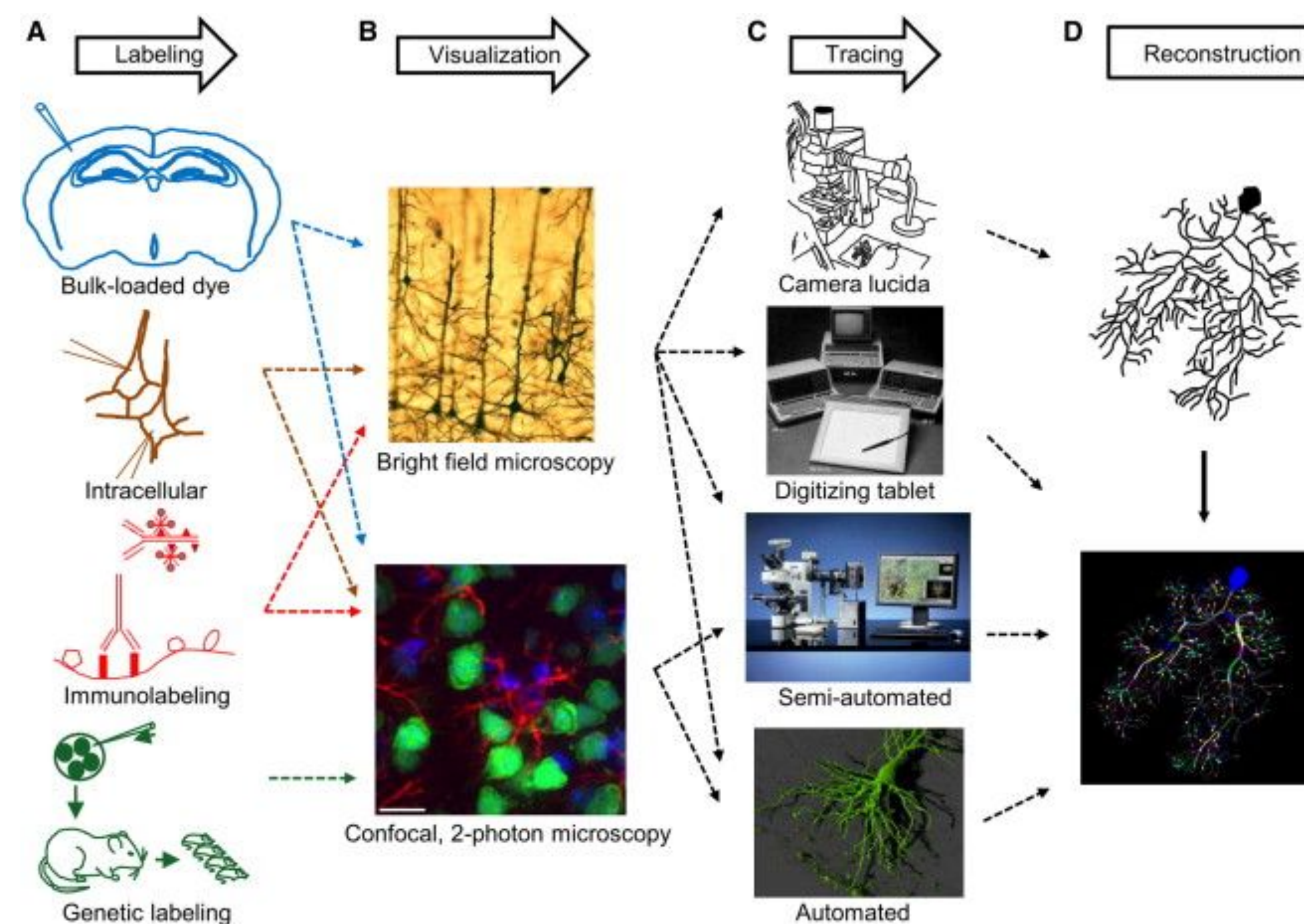
Computational Neuroscience: Tracing Neurons from the Fruit Fly Larva

Fatgezim Bela, Sumit Nanda, Giorgio A. Ascoli | fbela@gmu.edu,

Centers for Neural Informatics, Structures & Plasticity, George Mason University

ABSTRACT

My research project, details both the importance and procedure required to trace both simple as well as highly complex neurons. These neurons are also classified based of the extent and complexity of their branches.



From Nervous Systems to Digital Reconstructions of Neuronal Morphology¹

INTRODUCTION

Neurons morphology has been studied for the last 100 years. The most famous studier of neuromorphology was Santiago Ramón y Cajal. He used the Golgi method to trace neurons in culture. With technological advancements, we are now able to semi-automatically trace these structures with extreme precision and accuracy. The benefits of this have yet to be completely exploited. However, with the help of neuromorpho.org we hope to see the full potential for neuronal reconstructions realized.

RESEARCH QUESTION/HYPOTHESIS

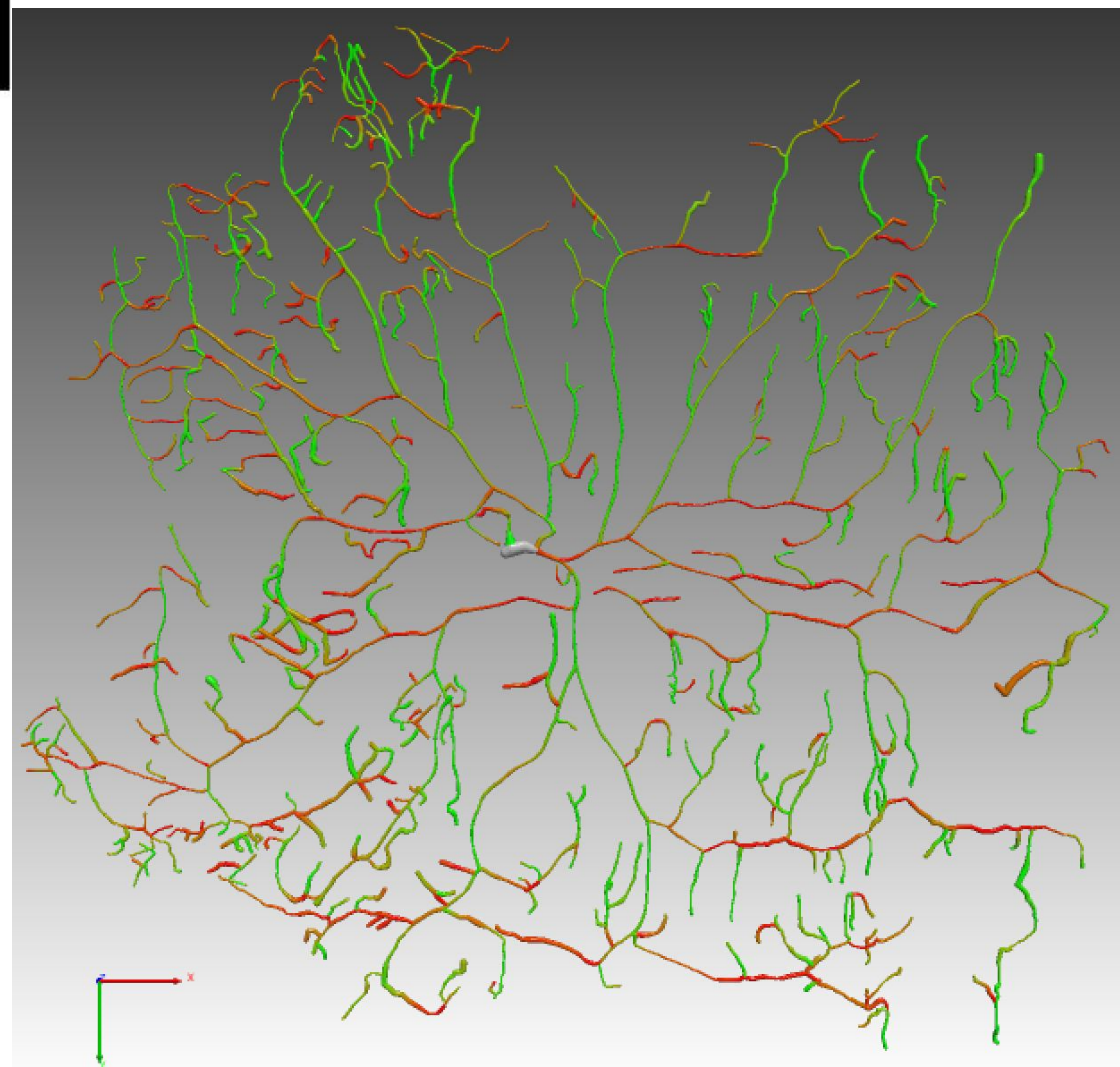
Have you ever wondered how 3D structures of neurons are reconstructed from microscopic image-stacks and how they are useful? With help from high quality neuronal reconstructions we will be able to determine growth patterns for fruit fly larva neurons.

METHODOLOGY

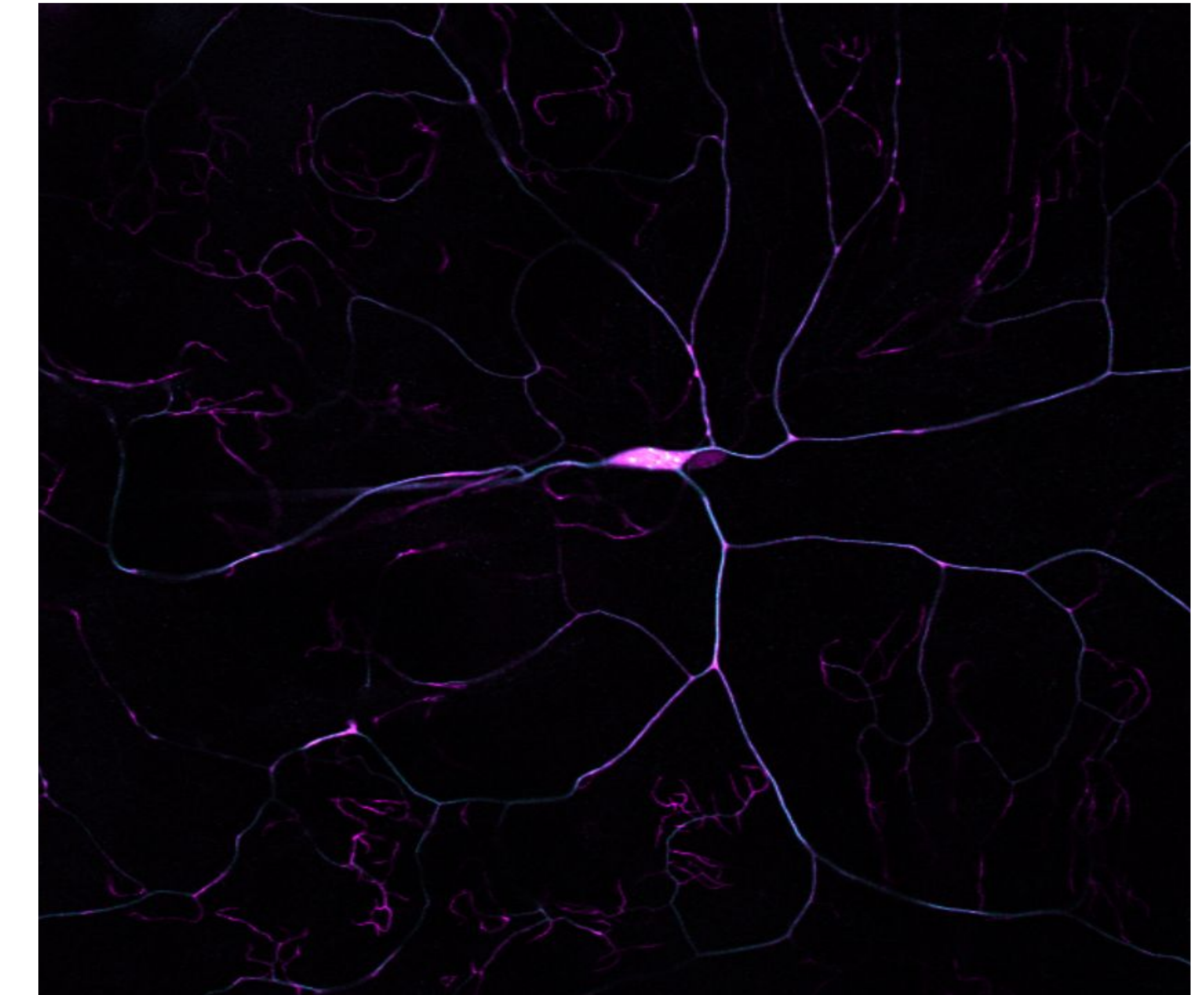
We take two or more live overlapping images of a fluorescently labelled neuron from Drosophila Larva and first process them in Fiji (Image J). In Fiji we process the two-channel images by splitting the image channels and creating an artificial membrane by adding the signals from the two. These images are then saved as Vaa3D raw file and transferred to Vaa3D where the parts of the same neuron are stitched together. Finally, the stitched image can be traced and the neuron can be effectively reconstructed into a digital vectorized tree structure in NeuTube. Once a neuron are fully traced, we will be able to compare it to the existing library of neurons and determine the patterns of its neural growth.

RESULTS

The results are very high quality reconstructions of neurons from different animals and parts of the cortex. These images will then be uploaded to Neuromorpho.org where other researchers can run various correlational or analytical analysis test on the neurons.



Reconstruction of Class 4 Neuron



Class 4 Neuron before reconstruction

DISCUSSION/CONCLUSIONS

While I currently am not applying any of the uses of the morphology data collected. I hope in the near future to use this data to be able to determine growth patterns in Fruit Flies. I would also like to test Sodium Chlorate's effect on E17 mice spinal cord axons. The applications of these studies could lead to a discovery of how to reverse or stop the effect of CSPG's which have been repeatedly proven to prevent the regeneration of axons.

ACKNOWLEDGEMENTS

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1. Parekh, Ruchi, and Giorgio A. Ascoli. "Neuronal Morphology Goes Digital: A Research Hub For Cellular And System Neuroscience". *Scimedirect.com*. N.p., 2017. Web. 20 Apr. 2017.